ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/358013413

Smart Greenhouse Monitoring System Using Internet of Things and Artificial Intelligence

Article *in* Wireless Personal Communications · June 2022

DOI: 10.1007/s11277-022-09528-x

CITATIONS
62

READS
15,129

6 authors, including:

Sultana Jahan Soheli
Noakhali Science and Technology University
17 PUBLICATIONS **141** CITATIONS

SEE PROFILE

Nusrat Jahan
Technical University of Darmstadt
2 PUBLICATIONS **62** CITATIONS

SEE PROFILE

Md Bipul Hossain
University of South Alabama
28 PUBLICATIONS **226** CITATIONS

SEE PROFILE

Apurba Adhikary
Kyung Hee University
108 PUBLICATIONS **845** CITATIONS

SEE PROFILE

All content following this page was uploaded by Md Bipul Hossain on 24 October 2024.

The user has requested enhancement of the downloaded file.

Wireless Personal Communications
https://doi.org/10.1007/s11277-022-09528-x

Smart Greenhouse Monitoring System Using Internet of Things and Artificial Intelligence

Sultana Jahan Soheli[1] · Nusrat Jahan[1] · Md. Bipul Hossain[1] · Apurba Adhikary[1,2] · Ashikur Rahman Khan[1] · M. Wahiduzzaman[3]

Accepted: 6 January 2022
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract
Climate change has already proven its terrible effect on agriculture. Although greenhouse is already an established system for crop production, with technological advancement it is possible to apply automation in many parts of this greenhouse. Therefore, an automated smart greenhouse based on an adaptive neuro fuzzy inference system (ANFIS) and Internet of Things (IoT) could be the best solution to boost the crop production inside the house. Where, four kind of weather data such as temperature, humidity, sunlight and soil-moisture are being collected by using sensors in real time. These collected data are then feed as input variables to the fuzzy control system. The fuzzy control system manipulate the data and ANFIS then make prediction for optimum values of the weather parameters. Thus farmers can monitor all the data and can decide the best value for temperature and humidity. The end users (farmers) can visualize all the data by a simple mobile app installed on their cell phone. GSM or TCP/IP is being used for all kind of data transferring. The FIS node also utilizes same networks to transfer IoT perception layer data to application layer. To ensure the data security, four types of potential IoT perceptron layer attacks are considered and shown their probability to occur through the confusion matrix. Later, necessary steps are taken to prohibit the attacks. Here winter crops are considered in the final simulation, when the optimum temperature in winter is 24º Celsius and humidity is 76.00%. The system is 93.62% capable to detect any attack or security breach at perception layer with a Precision value of 0.83, recall of 0.78 and FI score is 0.81. In comparison to other recently proposed and available systems, this work also combines IoT technology for identifying data threat on a network transfer with fuzzy set. This approach improves learning efficiency, improves prediction accuracy, and proved to be a feasible and effective automated greenhouse maintenance system. Simultaneously, the data collecting module and presentation schema of data from various sensors, as well as the security subsystem module, achieve cloud data storage and format conversion that is compliant with protocol format data. As a result, it may provide data traceability and durability for customized indoor agriculture quality and safety. Thus this modern greenhouse maintenance system is efficient, cost effective, secure and easy to use.

Extended author information available on the last page of the article

Published online: 21 January 2022

Keywords Internet-of-Things (IoT) · Smart greenhouse · Simulink model · Fuzzy Inference System (FIS) · IoT perception layer attacks · Wireless Sensor Network (WSN) · Confusion matrix

1 Introduction

A greenhouse is a surrounded structure that provides plants suitable ecological conditions for growing. Greenhouse crop cultivation systems are situated throughout the globe within a wide variety of climatic conditions in order to extend the production season of plants, where usually they will not grow well [1]. A Smart greenhouse system based on IoT and fuzzy inference system is a revolution in agriculture, creating an automatic microclimatic condition that optimize the conditions of plants growth through the use of IoT sensors and actuators that automate the controlling of weather factors enhance the plant growing process. In this system, by using different sensors farmers can remotely manage the greenhouse conditions and increased efficiencies during the seasonal performance and improved efficiency of water [2, 3]. There are so many problems that must be solved to boost the production of crops; greenhouse is an effective solution for that. But there are curtail facts that hinder the production inside a greenhouse. The more technologies one can implement to automate the operations of greenhouse the more the production. One of the most challenging problems is because of the inconsistent circulation of rainwater, it is very tough to the farmer to manage the water equally to all the crops on the whole farm. It requires some monitoring method that suitable for any weather condition, soil types and variety of crops. A smart greenhouse is the best solution for all of these problems. A farmer needs a continuous tracing of a greenhouse to maintain all environmental conditions that needed for different crops. If there is any change, the system should change or maintain that changes according to farmer instructions. Technological interconnection provides a transition of sensitive information which must be kept as confidential as possible. So, the security is very critical concern in IoT. The most basic level in IoT architecture is the perceptual layer (also known as recognition layer), which collects all kinds of information through networking equipment that contain RFID reader for all sensors and identifies the communicating network. The data refers object properties and environmental condition of the greenhouse. On the basis of these, this work solved the sensor's data security issues in different IoT perception layers [4, 5]. Concerning Internet of Things, different implementation methods and testbed development strategies are tested which are related to the system [6].

In this work, we introduced an IoT based smart greenhouse monitoring system. Where a procedure has been designed which will monitor and control present ecological conditions inside the greenhouse like temperature, humidity and soil moisture by using sensors those are connected by a WSN (Wireless Sensor Network) and all these data collectively sends to the farmer by using IoT and Wireless Communication. This system also controls factors such as heating, cooling, humidity inside the greenhouse by the use of fuzzy method. Data collected through wireless and IoT sensors are used as input to the fuzzy inference system. The result shows the accuracy of the model in decision making and provide quick access to crops growth to gain a satisfactory performance. According to the requirement of the crops the threshold value has been set, if any environmental conditions like temperature, soil conditions and humidity go below or above the threshold value, then sensor senses the changing in parameters and all the data will be transmitted to farmers, accordingly farmer(s) will take

the controlling decision and send to the system. We also investigated possible attacks to the perception layer and security threats to the IoT parts and proposes related solution through technologies.

G. Naveen et al. [7] used a thermal imaging for irrigation in the crop field. In addition to this, Image processing and data analytic techniques were used in thermal imaging for reducing crop water stress. Also, Cloud of things (CoT) network which comprises of Internet of Things and cyber physical system is used. Shouying L. et al. [8] proposed an intelligent greenhouse cultivation system which not only liberate the farmers' labor force and also worked on constraints on disastrous weather shock, pest outbreak and other undesirable environment limitations. Those systems have been put into practice in southeastern China. Özlem A. et al. [9] measured climate variables including temperature, relative humidity, soil moisture and light intensity by several sensors more efficiently and economically in the greenhouse. Moreover, data obtained from the sensors have been monitored and controlled with fuzzy-based control system. D.O.Shirsath et al. [10] implemented a design with Arduino platform for greenhouse monitoring, controlling temperature and soil moisture with the help of Web server using IOT. Suma et al. [11] described a GPS based smart monitoring system through the use of various sensors like temperature sensor, moisture sensor and PIR sensor. The data collected from these sensors are connected to the microcontroller through RS232. After the microcontroller is started, automatically an alert must be sent to the user by sending a message to the user through the GSM module. Chiyurl et al. [12] proposed a smart farm wired /wireless system with node server database and external control structure. A communication method was implemented using Bluetooth and LPWAN module in order to solve the power problem and space limitation of the wired system. Pavankumar et al. [13] measured various parameters of the soil using various sensors like temperature, humidity, soil moisture sensors while these sensed parameters and motor status will be displayed on user android application. Rodríguez et al. [14] developed a dynamic greenhouse environment simulator to understand the dynamic behavior of greenhouse environments. In addition, for allowing the selection of the greenhouse design, a web-based application was implemented. Dhamakale et al. [15] designed a Green House Climate Controller (GHCC) based on fuzzy logic programming which controls parameters such as temperature, humidity, & intensity of greenhouse. Ami et al. [16] proposed a cost effective IoT based intelligent greenhouse system and a cloud server to access the greenhouse parameters remotely without any human intervention. lmusaylim Z.A. and Zaman N. provided reviews regarding smart household appliances and working procedures of various sensors, their present state and challenges which are helpful for this work as well for a context-awareness internet of things (IoT) [17]. Performance analysis of data transmission rate and decoding rate using minimum energy channel codes for nanoscale wireless communication related to this work implanted according the research of Adhikary A. et al. [18]. Srivastava A. and Das D.K. demonstrated a comprehensive review on the application of Internet of Thing (IoT) in smart agriculture which lacks an individual system development, however the reviewed some already installed systems [19]. Feng X. et al. proposed only the working of wireless communication technologies on internet of things for precision agriculture [20]. A major issue in most of above mentioned article is that these systems cannot detect the possible attacks of perception layer and security threats to the IoT and being unable to calculate the accuracy of the model in decision making.

2 Methodology and Implementation

2.1 System Outline

2.1.1 Basic System Breakdown

The greenhouse system consists of a data collection unit, data communication unit, and data processing unit and system control section. Figure 1 depicts the system by a detailed flow chart with primary parts from where the pseudo code of the system become more clarified.

- Data Collection Unit: The continuous environmental variables such as ambient temperature, humidity, soil moisture, and light intensity inside the greenhouse are measured by sensors. Here we used DHT11 as temperature sensor, YL-69 as soil-moisture sensor and light-dependent resistor LDR sensor for sensing the sunlight intensity. All these sensors periodically collect real time environment conditions and send the data to ANFIS.
- Data Communication Unit: To send and receive data wirelessly between sensors and Arduino UNO (ATmega328P microcontroller), MQTT (Message Queuing Telemetry

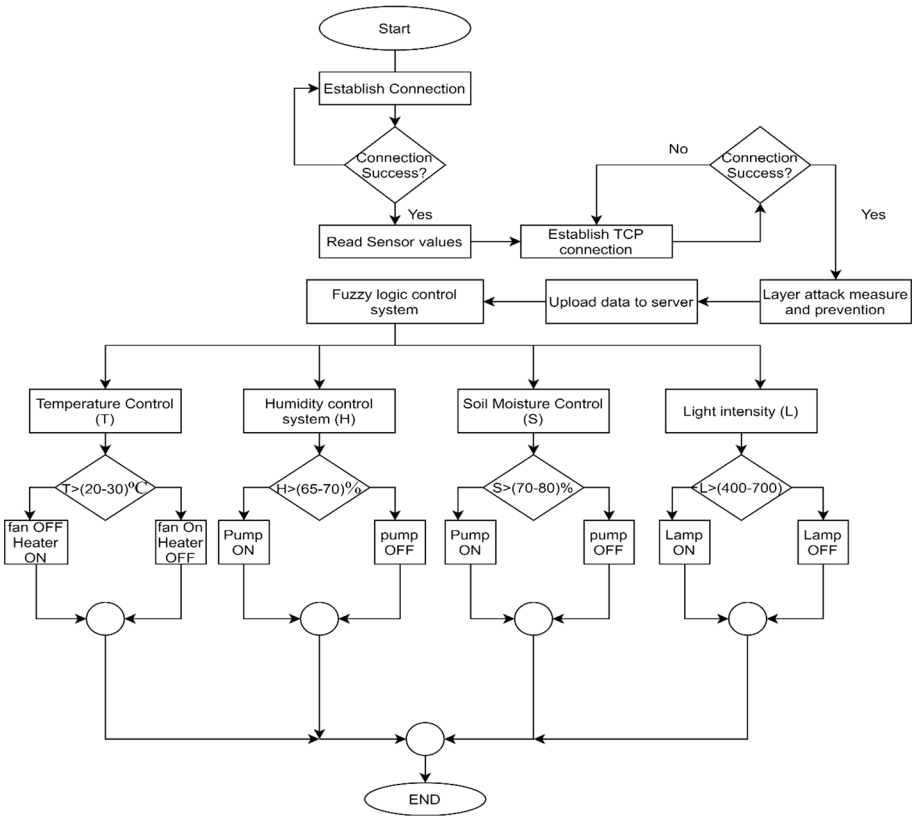


Fig. 1 Flow chart of greenhouse monitoring and control system

Transport) protocol is used. The connection between Arduino and other mobile devices is established by GSM or TCP/IP.

- Data processing: Fuzzy logic controller receives input from sensors and to produce output, the sensor data is calculated for regulating greenhouse parameters to make the intelligent decision for environment.
- Control section: The control section involves actuators such as fan, heater, humidifier, motor, exhaust fan which responses according to the sensor data. FLC controls the heating, fan, humidification, etc. by adding new sets and rules in ANFIS it is possible to adjust the temperature, humidity, soil moisture, and light intensity of greenhouse based on data processing section status.

2.1.2 Pseudo Code Algorithm for the Approach

Start the system.
(2) Establish the necessary connections with all the sensors and processing modules.
(3) If, the connection establishments are successful, read parameters' values from sensors and turn on the TCP connection.
(4) Else check out all the connections and sensors.
(5) (i) upload the greenhouse data to the cloud for Fuzzy Logic Control System implementation.
(ii) Start the IoT layers attack(s) detection, countermeasure and prevention system.
(6) Feed the fuzzy control system with temperature, humidity, moisture and light intensity parameters values for calculating the optimum temperature and conditions.
(7) According to the parameter values take decision about fan, heater, pump and lamp.
(8) End.

2.2 Working Procedure of Different Sensor

2.2.1 Working Procedure of Temperature Sensor

DHT11(Fig. 2) temperature-humidity sensor has been used here to measure the surrounding temperature and air in every two hours interval in day time and with a four hours interval in night time. As this type of sensors connects to a high- performance 8-bit microcontroller, it then send the data to FIS through the microcontroller. During the experiment, we observed that the greenhouse has some static humidity (65–70%) and temperature levels (20–30ºC). As it was winter. Above/below those levels, microcontroller sends corresponding signals to the interface device [21]. The micro- controller is programmed in a way that once the temperature is not up to the desired temperature

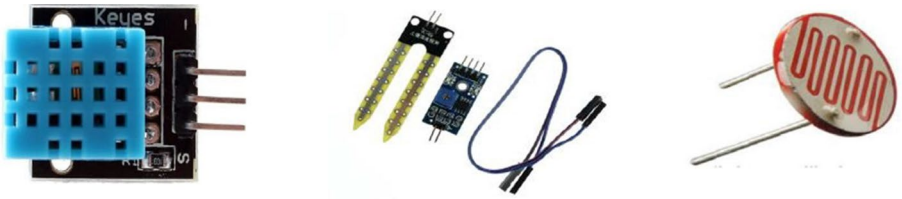


Fig. 2 (from left to right) DHT11 light sensor, YL69 moisture and humidity sensor and LDR light intensity sensor

such as below 20ºC, the bulb (that represents heater) can activate. And, if the temperature is found to be beyond the desired temperature (above 30ºC) then the cooling fan (that represents air conditioner) would activate. DHT11 detector takes analog input from the humidness and temperature sensors and provides digital output [21]. Flowchart of temperature sensing is shown in Fig. 3a. For control, if the greenhouse inside temperature rises above a threshold value then the actuator (Fan) will be automatically switched on to lessen the temperature. When the temperature reaches to a threshold value according to Table 1 then the cooling fan will be automatically switched off. If the temperature decreases below the desired temperature, a heater will be switched on to increase the temperature and when the temperature reaches to desired range, the heater will be automatically stopped working.

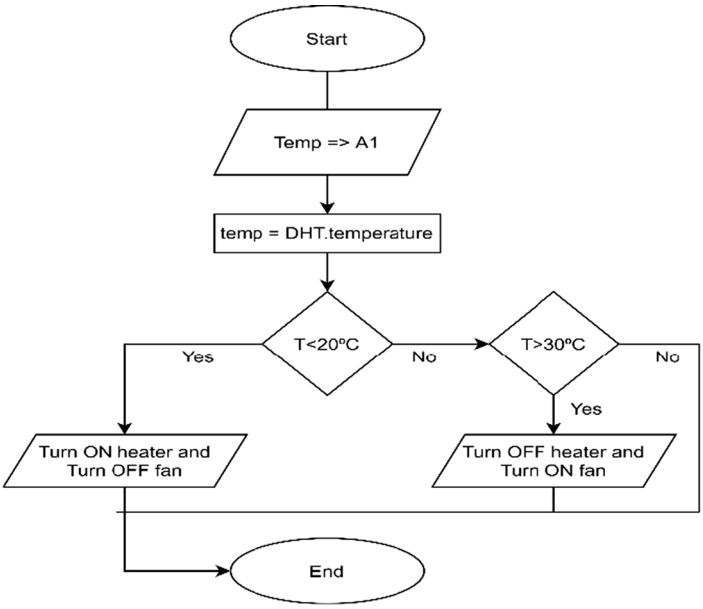
2.2.2 Working Procedure of Humidity Sensor

A humidity sensor or hygrometer, measures surrounding humidity of the environment and periodically reports to the control system. If the humidity rises above the required threshold value (65–70%) then the DHT11 temperature- humidity sensor (Fig. 4b) will detect it and Arduino will be programmed to act accordingly. When the humidity of the system increases above (> 70%) the desired value, the exhaust fan is programmed to turn on which derive the extra humid air out of the system. And when the air is less humid (below 65%RH) than the given humidity, the fan will turn on. (Flow chart is shown in Fig. 3b).

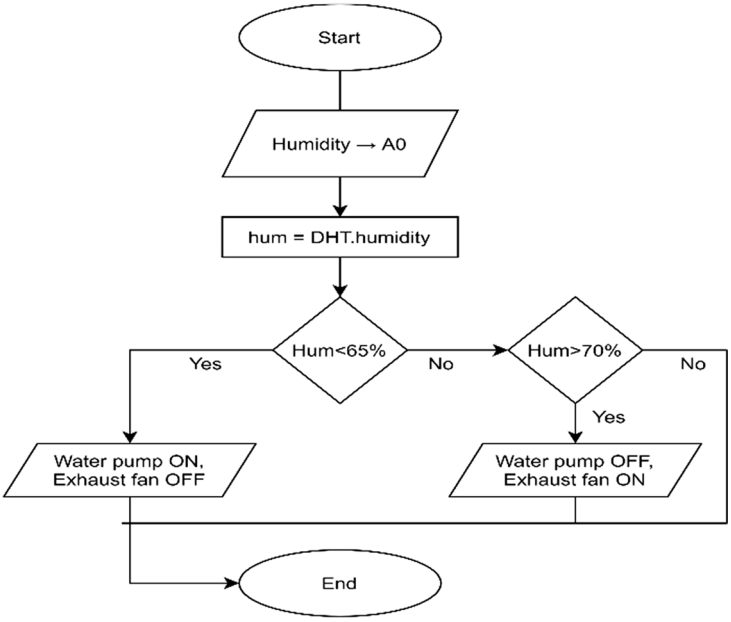
For air humidity control, the humidity sensor will monitor the humidity according to Table 1, Humidity control part and an actuator (e.g. exhaust fan, water pump) will be activated as per the requirements. Humidity sensor will measure the humidity of the air in the greenhouse in terms of percentage Relative Humidity (%RH) [16]. When humidity level exceeds the threshold level the exhaust fan will be switched on to maintain the appropriate climatic conditions for the greenhouse plant.

2.2.3 Working Procedure of Soil-Moisture Sensor

Soil moisture sensor measures the volumetric water potential in soil. It is used to measure the loss of moisture over time due to evaporation for various species of plants, monitor soil moisture content to control irrigation in greenhouses. In this research, YL-69 soil moisture sensor (Fig. 2) has been used to remind the user to water greenhouse plants and monitor the moisture content in the soil. Microcontroller sends the signal of soil condition to the corresponding devices to turn on/off the pump to meet the requirements (70–80%). Moreover, once the soil has wetness to a threshold level the microcontroller sends signal to switch off the pump through interfacing devices reliable with the sensors output (Fig. 3c). For controlling greenhouse soil moisture (see Table 1, Soil-Moisture Part), a soil moisture sensor is used which constantly measures the water present in the soil. If there is no suitable volume of water in the soil (< 70%) then the water pump starts automatically to provide water until threshold value is meet. When moisture level reaches to the desired level, pump will automatically stop working.

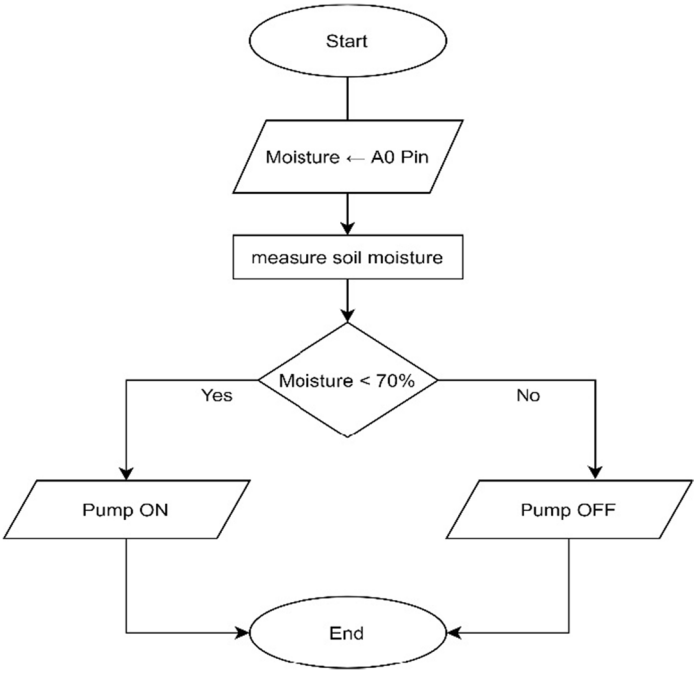


(a) Flowchart for temperature measurement

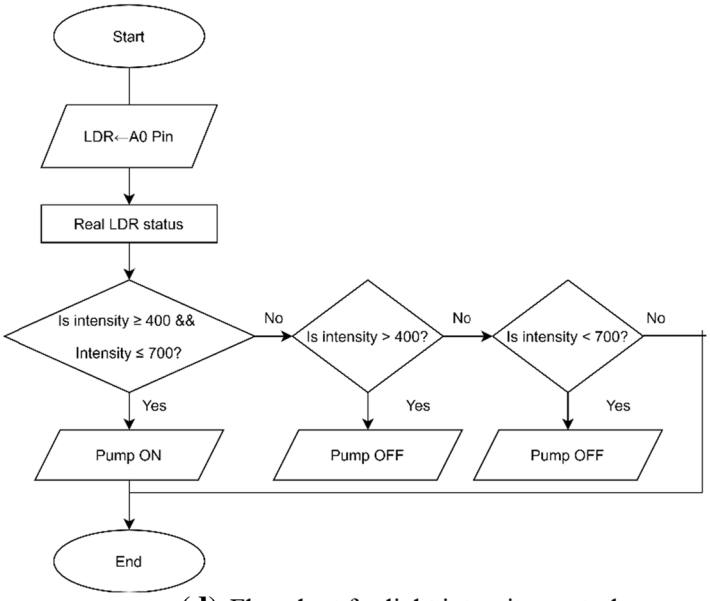


(b) Flowchart for humidity measurement

Fig. 3 **a** Flowchart for temperature measurement **b** Flowchart for humidity measurement **c** Flowchart for water motor control **d** Flowchart for light intensity control



(c) Flowchart for water motor control



(d) Flowchart for light intensity control

Fig. 3 (continued)

Table 1 Controlling factors and threshold values for all parameters

Temperature controlling			Humidity control			Soil-moisture control		Light intensity control	
Temperature value	Fan	Heater	Humidity (In %Rh)	Exhaust fan	Water pump	Soil-moisture value	Pump	Ldr sensor readings	Lamp condition
20–30 ºC	OFF	OFF	65–70	OFF	OFF	70–80%	OFF	400–700	OFF
< 20	OFF	ON	< 65	OFF	ON	Below 70%	ON	< 400	ON
> 30	ON	OFF	> 70	ON	OFF	Above 80%	OFF	> 700	OFF
Note: Threshold value (20–30) ºC			Note: Threshold value (65–70) %RH			Note: Threshold value (70–80) %		Note: Threshold Value (400–700)	

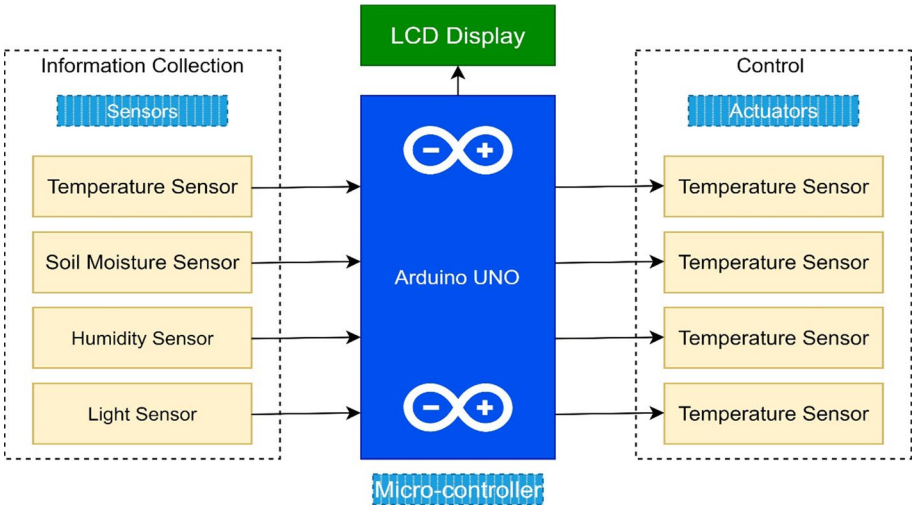


Fig. 4 Block Diagram of greenhouse monitor and control system

2.2.4 Working Procedure of Light Dependent Resistor (LDR) Sensor

A Light Dependent Resistor (LDR) sensor (Fig. 2) also known as photo resistor is an active component that decreases resistance with respect to receiving luminosity (light) on the component's sensitive surface. If light is absence in greenhouse the detector detects it and sends signal to the microcontroller to set an optimum light condition for the plants. Additionally, when light intensity is at a desired level (400–700) there is no change in the light condition according to Fig. 3d.

For controlling light intensity, the sensor senses the light intensity inside the greenhouse environment according to Table 1. If there is inadequate light in the greenhouse, the light will be switched on by the light sensor and light will be switched off if the plants in the greenhouse get adequate light.

2.3 System Simulation

PROTEUS has been used to test programs and embedded designs for electronics before actual hardware testing. Detailed are at Table 2. Where the careful consideration for data routing in case Internet of Things are made followed by rigorous tests with real time external environmental parameters [22].

In this interfacing, DHT11 sensors library (DHT11, DHT21 and DHT22) from Adafruit is used. This library initiates the DHT11 sensor and reads the values of the humidity and temperature. Simulation of the circuit is completed in Arduino and checked it at before and after run. And Table 3 depicts the data chart of simulation system.

Table 2 All connection setup

Connection setup	
Soil moisture sensor (yl-69)	Temperature and humidity sensor
Step 1: The soil moisture sensor (YL-69) consists of three pins namely A0, D0, VCC and SGND	Step 1: The temperature and humidity sensor (DHT11) consists of positive, output and negative pins
The A0 pin of soil moisture sensor (YL-69) is connected to the analog input pin A0 of Arduino UNO	The positive pin of DHT 11 is connected to the 5 V power supply of arduino UNO
Connect VCC of YL-69 to 5 V power supply pin of arduino UNO	The output pin of DHT 11 is connected to the digital output pin D7 of arduino UNO
Connect the ground pins of YL-69 and arduino UNO	The negative pin of DHT 11 is connected to the GND pin of arduino UNO
Step 2: Compile the code to check errors	Step 2: Compile the code to check for errors
Step 3: Upload the code to the Arduino board	Step 3: Upload the code to the Arduino board

Table 3 Data chart of simulation system

SN	Time	Temperature	Humidity
1	2019–12-12 11.29.41	27.00	74.00%
2	2019–12-12 11.29.41	27.00	74.00%
3	2019–12-12 11.29.43	27.00	74.00%
4	2019–12-12 11.29.44	27.00	74.00%
5	2019–12-12 11.29.45	28.00	76.00%
6	2019–12-12 11.29.53	27.00	74.00%
7	2019–12-12 11.29.53	28.00	76.00%
8	2019–12-12 11.29.54	28.00	76.00%
9	2019–12-12 11.29.55	27.00	74.00%
10	2019–12-12 11.29.59	27.00	74.00%
11	2019–12-12 11.29.59	27.00	74.00%

2.4 Block Diagram of Greenhouse

The greenhouse system consists of data collection section and control section. Data collection section involves various sensors to monitor the greenhouse parameters and sends the data to the microcontroller (Arduino ATmega8233) which in turn displays the collected data to a display module. The control section of the microcontroller involves actuators such as fan, heater, humidifier, motor, exhaust fan which responses to the sensor data. The block diagram of the whole greenhouse system is shown in Fig. 4.

2.4.1 Control Section

Collected data transmitted to the Arduino board. The Arduino board consists of microcontroller ATMEGA328P which is responsible for controlling the switching on/off of the motor on which water sprinklers can be attached.

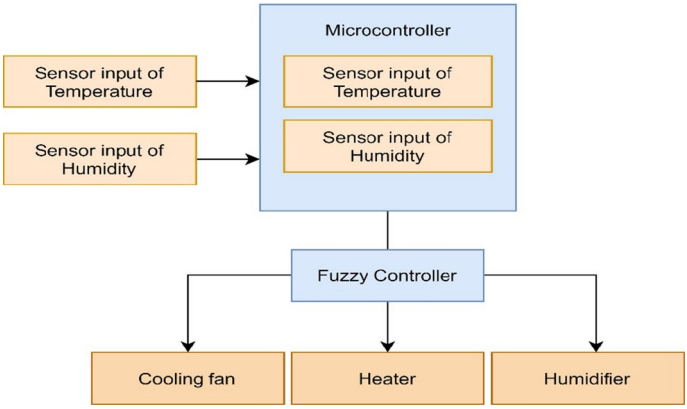


Fig. 5 Fuzzy logic process for controlling duration of heating and watering

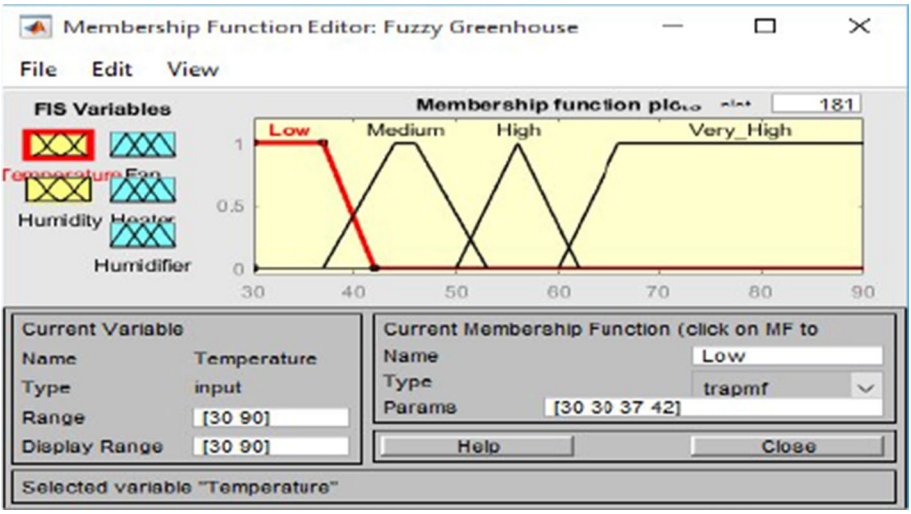


Fig. 6 Membership function specification for temperature

2.4.2 Fuzzy Inference System of Greenhouse

The fuzzy logic controller used for greenhouse control has four inputs and two outputs which drive the actuators related to humidifier, ventilation and heating to be switched on or off. The preliminary result of controller that embeds structural human knowledge about irrigation scheduling into an analytic fuzzy logic model (FLM), in order to provide quick access to the end users to a particular crop's production and growth-related variables, and allowing for future data driven decisions on the spot. The process for controlling the time duration of watering, heating in the greenhouse then decided by the farmers. The process is shown in Fig. 5.

Membership functions make the most fundamental variance between conventional logic and fuzzy logic. By using membership functions, the weights of the state and control variables are expressed continuously in the range of 0–1 [23]. Once the process of inferring,

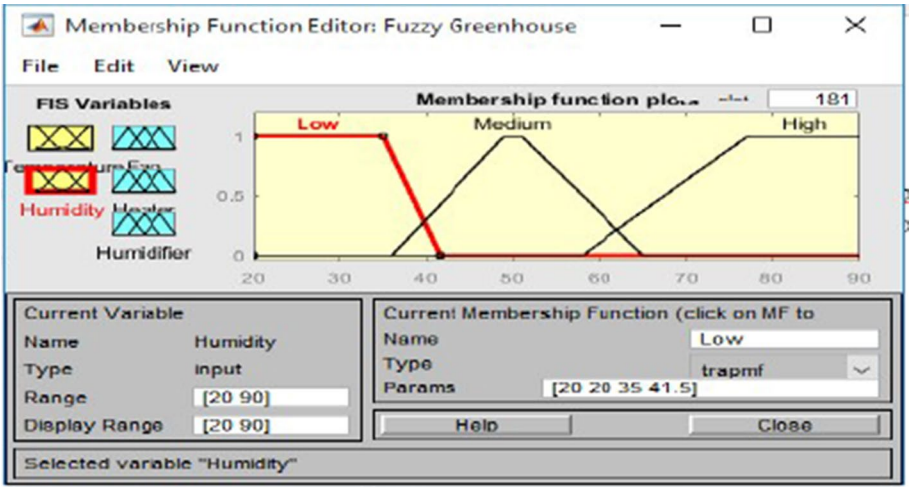


Fig. 7 Membership function specification for humidity

scaling and combining for functions are done, they are defuzzified into a crisp output which drives the actuators. The first input is temperature with conditions of low, medium, high, very high use trapmf using the membership function types, shown in Fig. 6. A second input is humidity with conditions of low, medium, high, very high use trapmf using the membership function types. The range values from 20 to 90 express the magnitude of the humidity in the greenhouse. Which is depicted at Fig. 7.The range values from 30 to 90 express the magnitude of the temperature in the green house (Fig. 6). Condition of control variables used: (a) Fan, (b) Heater and (c) Humidifier are shown in Fig. 8.

2.4.3 Fuzzy Logic-Based Threat Modeling

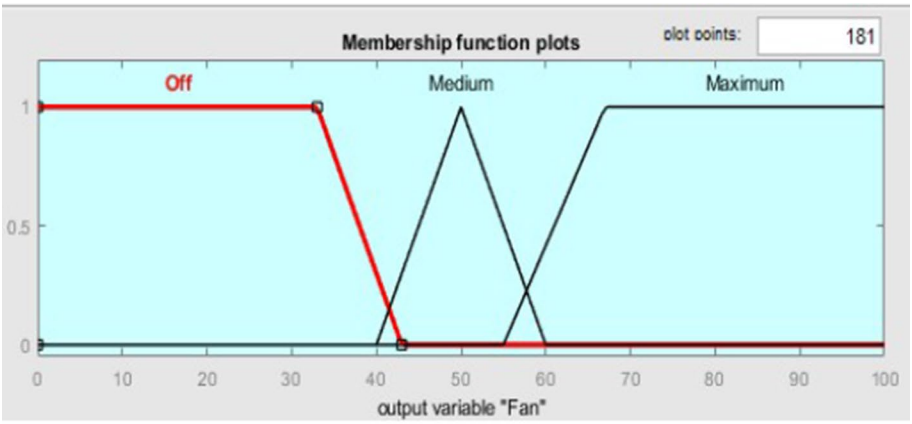
The system is designed using fuzzy inference system which is a popular computing framework based on the concept of fuzzy set theory, fuzzy if–then rules, and fuzzy reasoning, like Fig. 9. The steps involved in the design are:

$$\text{Fuzzification} \rightarrow \text{Rule Evaluation} \rightarrow \text{Defuzzification} \rightarrow \text{Output}$$

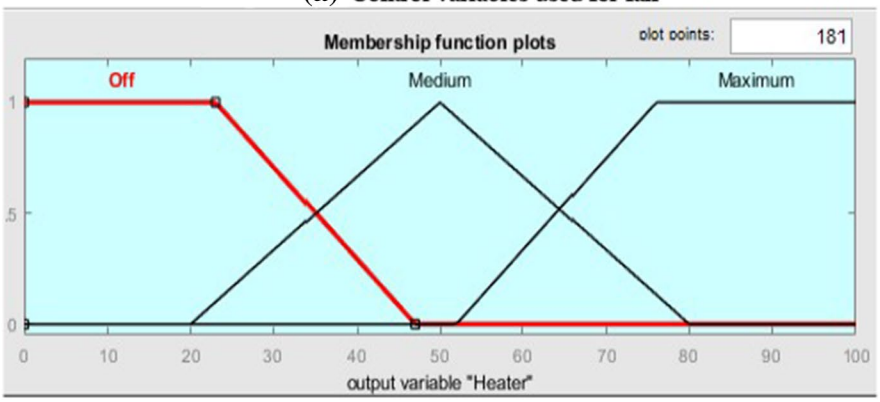
(1) *Fuzzification*: This is the method of creating membership values for fuzzy variables using membership functions. The initial stage is to take the crisp inputs information (Information Disclosure, Denial of Service and message modification) and determine the degree to which these inputs belong to each appropriate fuzzy set. Once the crisp inputs are obtained, they are fuzzified against the appropriate linguistic fuzzy sets.

(2) *Rule Evaluation* This is the next step where the fuzzified inputs are applied to the antecedents of the fuzzy rules. Since the fuzzy rule has multiple antecedents, fuzzy operator (AND or OR) is used to obtain a single number that represents the result of the antecedent evaluation. We apply the AND fuzzy operation (intersection) to evaluate the conjunction of the rule antecedents.

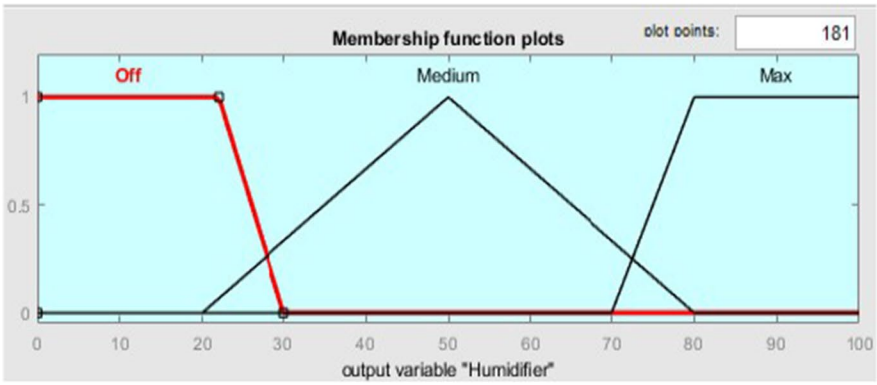
(3) *Defuzzification* This is the last step in the fuzzy inference process, which is the process of transforming a fuzzy output of a fuzzy inference system into a crisp output. This



(a) Control variables used for fan



(b) Control variables used for heater



(c) Control variables used for humidifier

Fig. 8 **a** Control variables used for fan **b** Control variables used for heater

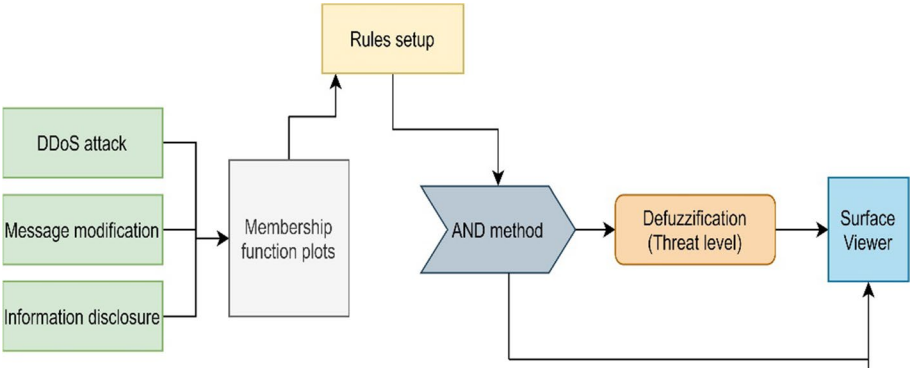


Fig. 9 Architecture for fuzzy logic-based threat modelling

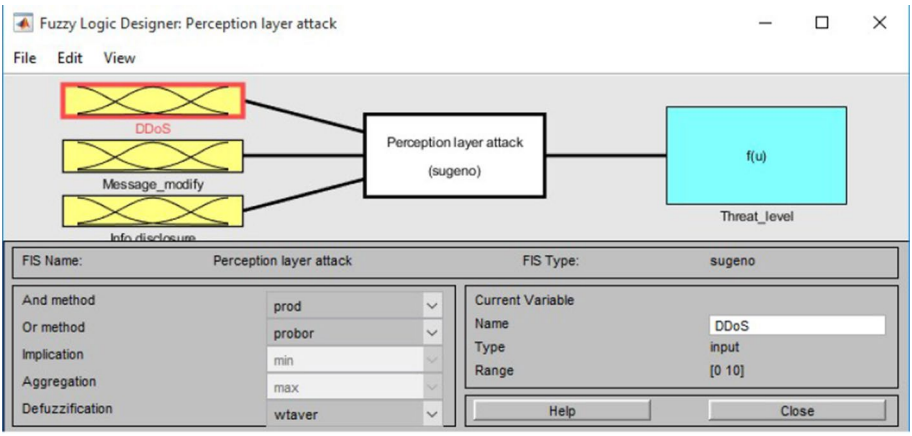
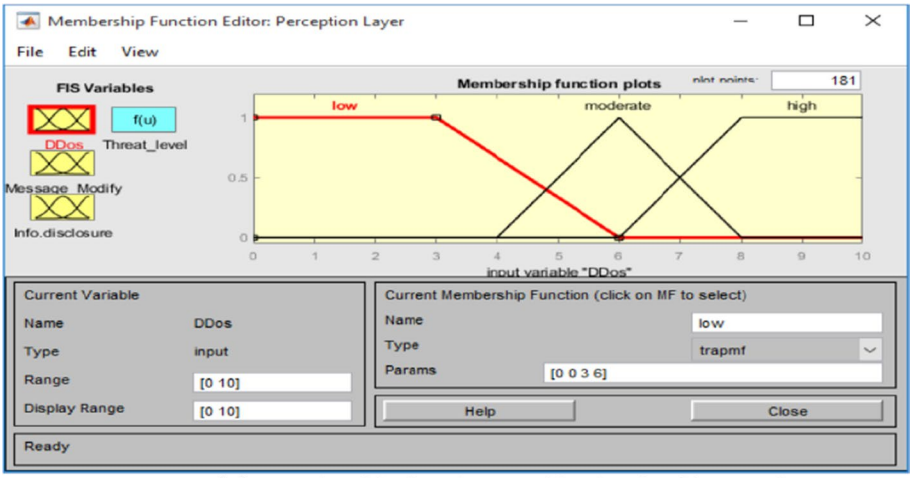


Fig. 10 Setting parameters for perception layer attack

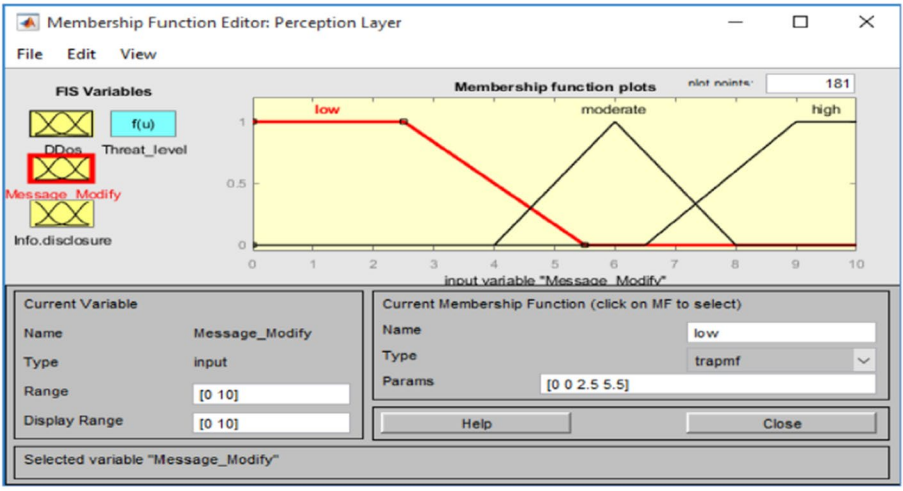
step was done using Centroid technique because it is most commonly used method of defuzzification.

(4) *Implementation of Threat modelling in FIS-2* Fig. 10 is showing the FIS functions. In this case, the chosen model is sugeno. Figure 11a, b, c are representing the membership function specification for DDoS attack, Membership function specification for message modification, Membership function specification for Information disclosure.

(5) *Rule Viewer* The rule viewer shows a graphical representation of each of the variables through all the rules, a representation of the combination of the rules, and a representation of the output from the defuzzification. It also shows the crisp value output of the system.



(a) Membership function specification for ddos attack

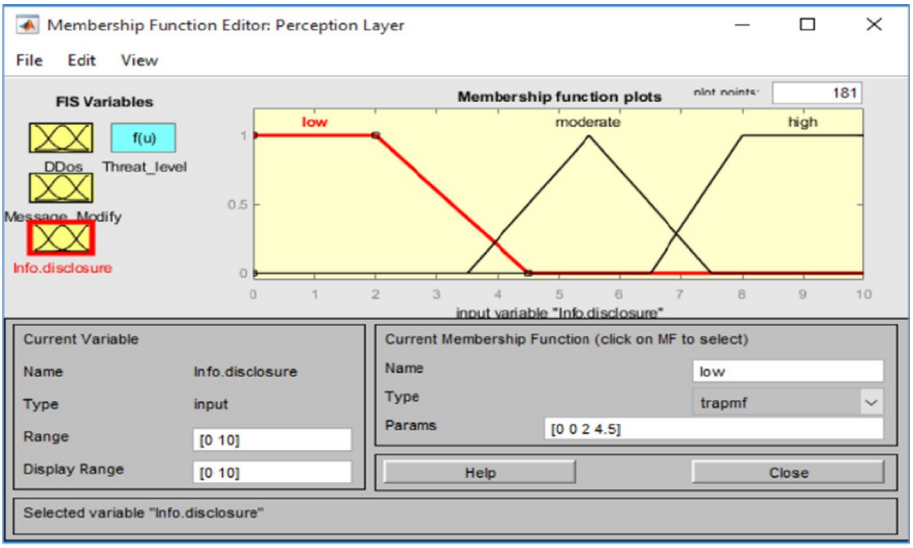


(b) Membership function specification for message modify attack

Fig. 11 **a** Membership function specification for ddos attack **b** Membership function specification for message modify attack **c** Membership function specification for information disclosure attack

2.4.4 Internet-of-Things and Security Issues

Internet of Things (IoT) is a type of network technology, which senses the information from different sensors and make anything to join the Internet to exchange information [24]. Utilizing IoT technology, any system can be controlled automatically from anywhere through the use of diverge sensors and actuators [25]. Sensors allow collecting data from environment which in turn allows to make intelligent decisions. The IoT input devices have various input modules attached that can sense what is going on in the environment (e.g., light, temperature, humidity, soil control, co_2 control) and be able to upload data into the cloud [26]. User can view sensor data with the help of IoT output device such as LED and a command



(c) Membership function specification for information disclosure attack

Fig. 11 (continued)

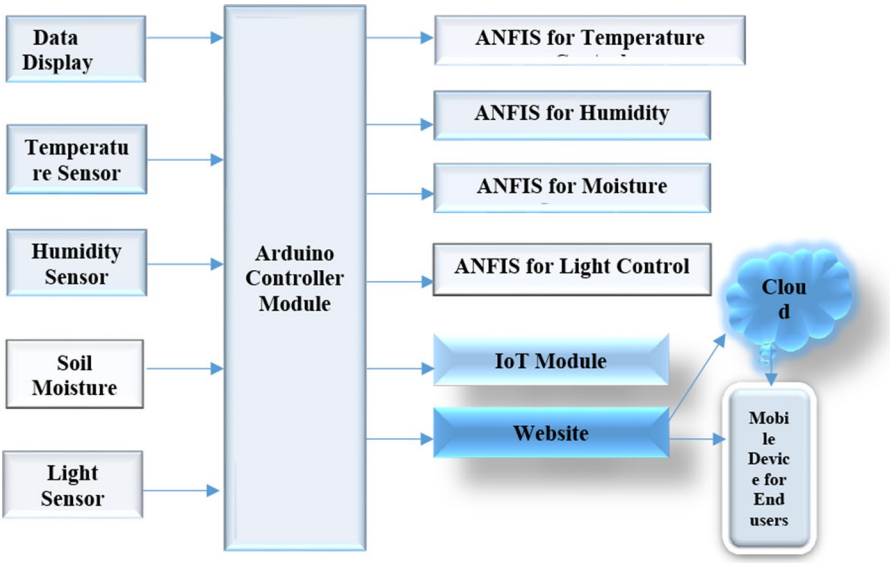


Fig. 12 Structure of proposed smart greenhouse

are sent to the actuator (fan, motor, pumps, heater, and humidifier) in response to the sensors data along with the whole system structure shown in Fig. 12. The data security of IOT perceptron layer should be furnished with the properties such as identification, confidentiality, integrality [27]. Different from internet, the IoT will be applied to the crucial areas. So

Table 4 Different attacks and their weight for finding the effects

Linguistic variable: DDoS attack		Linguistic variable: Information disclosure		Linguistic variable: Message modification		Linguistic variable: Threat level	
Linguistic value	Numerical range	Linguistic value	Numerical range	Linguistic value	Numerical range	Linguistic value	Numerical range
Low	[0, 0, 3, 6]	Low	[0, 0, 2, 4.5]	Low	[0, 0, 2.5, 5.5]	Low	[10, 20, 30, 40]
Moderate	[7, 9, 11]	Moderate	[3.5, 5.5, 7.5]	Moderate	[7, 9, 11]	Moderate	[20, 30, 50, 80]
High	[9, 11, 13]	High	[6.5, 8, 10, 10]	High	[6.5, 9, 10, 10]	High	[40,60, 80, 100]

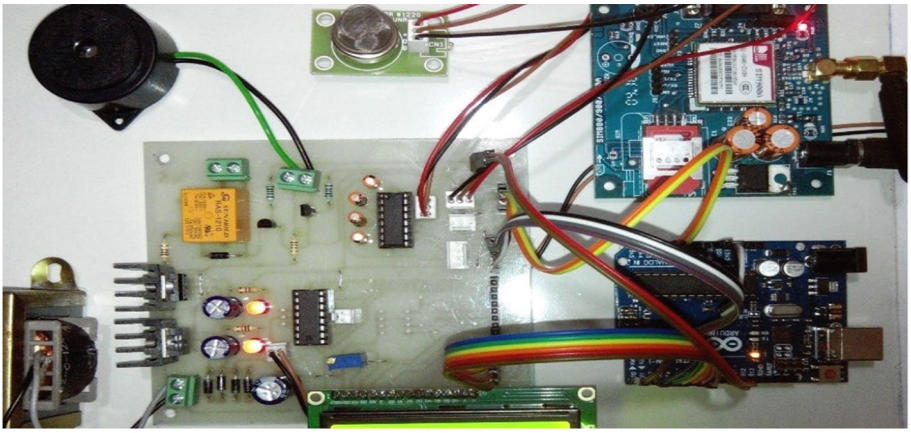


Fig. 13 System prototype

security needs in the IoT will be higher in availability and dependability. Linguistic variables are assigned for each threat of perceptron layer as Low, Moderate, and High while for threat level as Low, Medium, and High. For each input threat their values range from 0 to 10 while for output, ranges from 0 to 100. Table 4 contain the linguistic variables and their ranges.

3 Result Analysis & Discussion

Our system prototype with Arduino microcontroller is given in Fig. 13.

Costing Information of the system prototype The duration of this research work is almost 1 year. Total cost of this system was approximately 1500 USDAs this project required different types of devices, sensors, cloud service and other non-recurring engineering costs in order to make an efficient version for an implementation and test run. However, we tried to keep our expenses minimum by doing majority of works by ourselves.

3.1 Surrounding Air Temperature and Relative Humidity

Weather constraints such as humidity, temperature, and soil moisture, etc. have an effect on agriculture. Starting from seed sow to crop collection every step has different requirements of climatic parameters. To get ideal progress, it is essential to ensure appropriate temperature, humidity, light and soil moisture level. From our simulation, (Time: December2019/January 2020 and rechecked at January 2021) as data were taken at winter season the

Table 5 Temperature and Humidity Status

No	Temperature value	Action taken	Humidity value	Actuates
1	< 20	Turn on light	< 30	Turn on fan
2	> 35	Turn off light	> 75	Turn off fan
3	> 20 & < 35	Turn off light	> 30 & < 75	Turn off fan

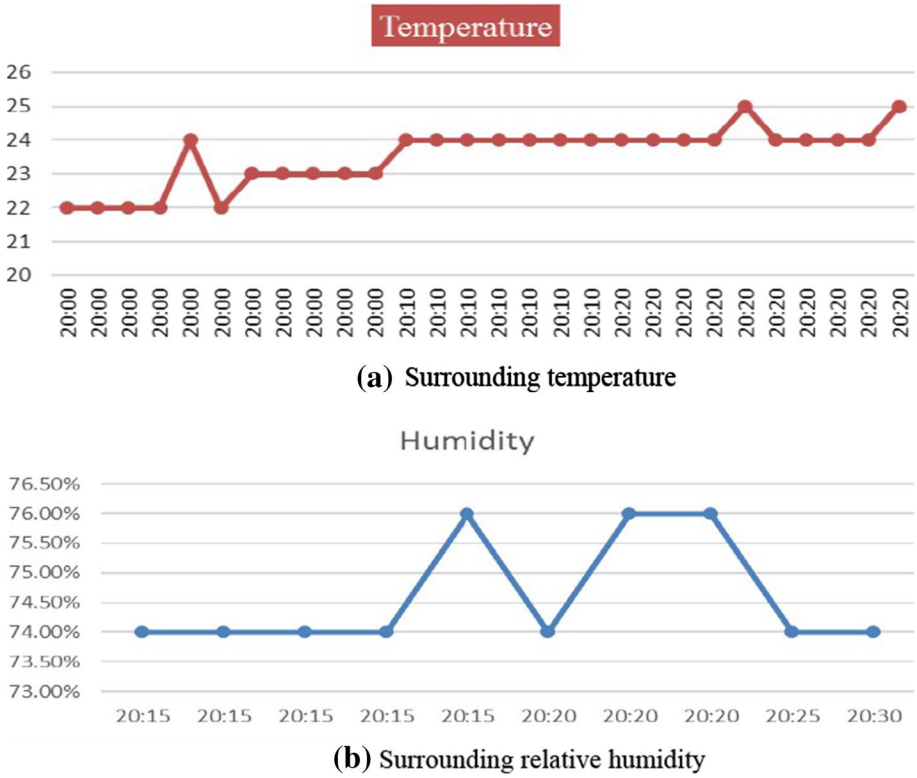
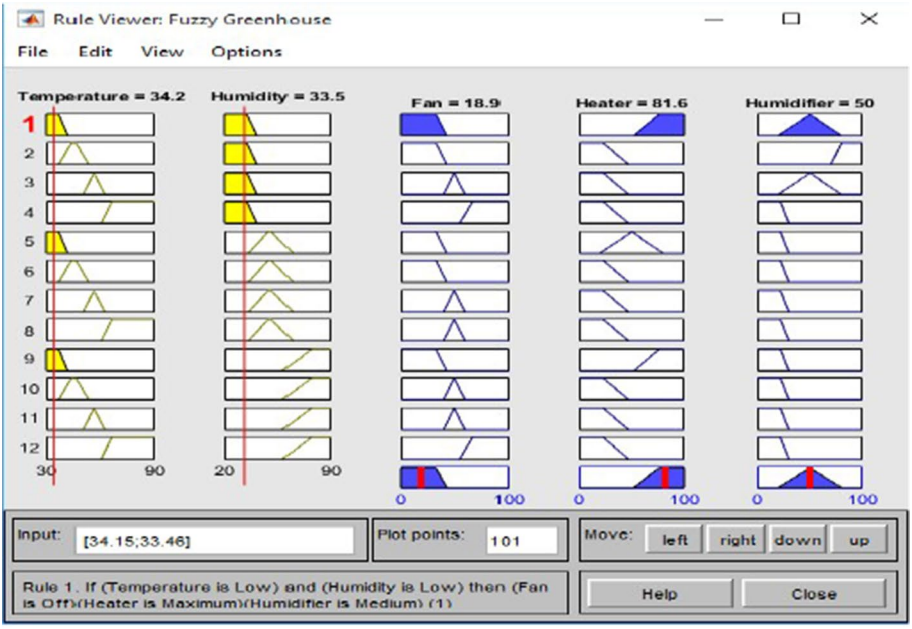


Fig. 14 **a** Surrounding temperature **b** Surrounding relative humidity

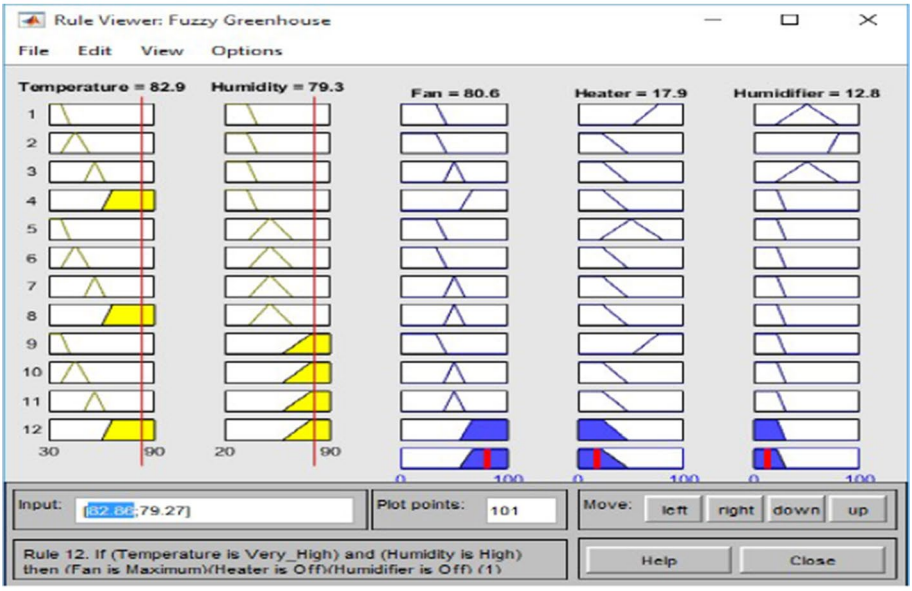
greenhouse can give best crop production if the parameters values could be like Table 5. The graphical illustration of the data sensed according to Table 5 at the simulation part is given in Fig. 14a and Fig. 14b.

3.2 Fuzzy Logic Results of Temperature and Humidity

For rule1, the input humidity has to be of any value in the range of 20 to 40% (This is "low" membership function for humidity input) while the input temperature has to be of any value in the range between 30 to 42℃ (This is "low" membership function for temperature input) while Fig. 15a shows the output results of these inputs. It can be seen that the output of fan is low, heater is high and humidifier is medium when the temperature is 34.2℃ ("low" membership function value of temperature input) and humidity was 33.5% ("low" membership function value of the humidity input). These outcomes satisfy rule 1. For rule 12 to be invoked, the input temperature has to be of any value in the range of 60 to 108℃ (This is "very high" membership function for temperature input) while the input humidity has to be of any value in the range of 58.2 to 116%. Figure 15b shows the output results at these inputs. It can be seen that the output of fan is "maximum", heater is "off" and humidifier is "off" when the temperature was 82.9℃ ("very high" membership function value of temperature input) and humidity was 79.3% ("high" membership function value of the humidity input). These outcomes satisfied design rule 12.



(a) Output results when rule 1 was invoked

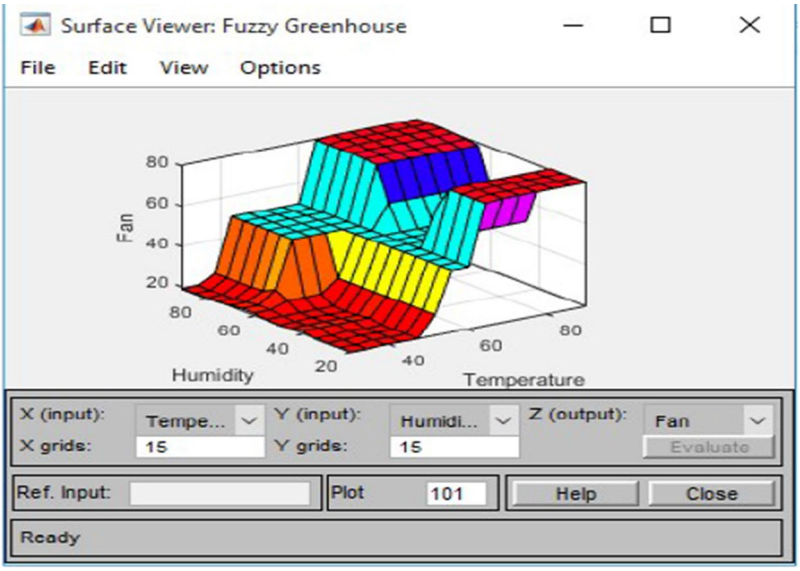


(b) Output results when rule 12 was invoked

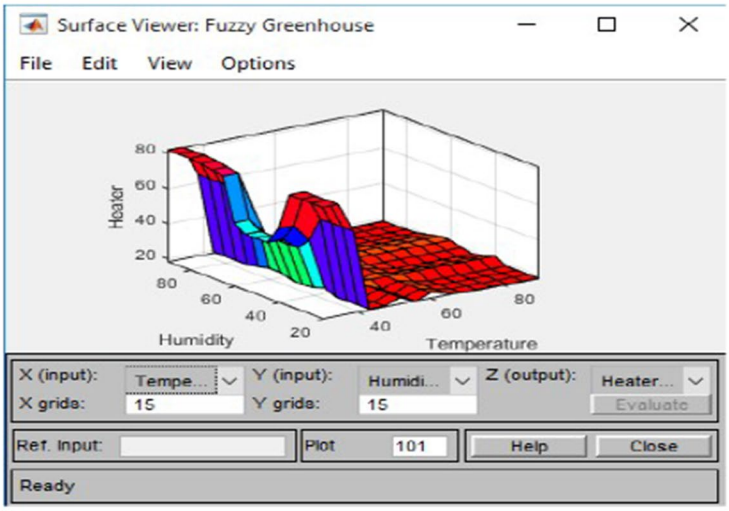
Fig. 15 **a** Output results when rule 1 was invoked **b** Output results when rule 12 was invoked

3.3 Results of Rule Viewer

The results of weather parameters rule viewer can be seen in the form of a three-dimensional surface view as shown in Fig. 16. The surface viewer is plotting a graph of the data

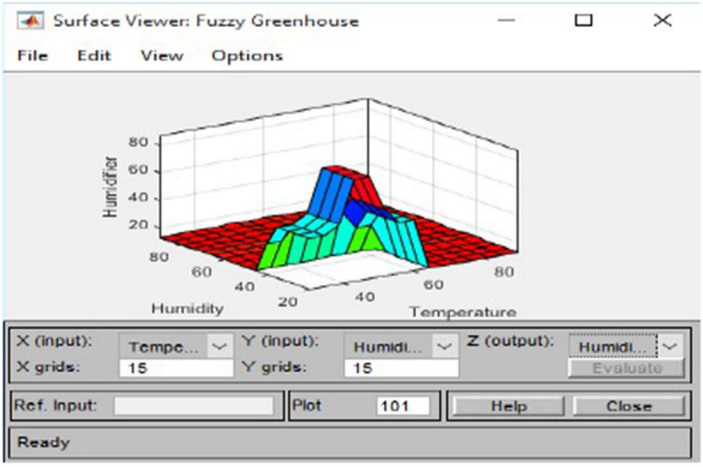


(a) Surface view z-output of temperature input X-axis and humidity input Y-axis



(b) Surface view z-output of temperature input X-axis and humidity input Y-axis

Fig. 16 **a** Surface view z-output of temperature input X-axis and humidity input Y-axis **b** Surface view z-output of temperature input X-axis and humidity input Y-axis **c** Surface view Z-Output of temperature input X-Axis and humidity input Y-Axis



(c) Surface view Z-Output of temperature input X-Axis and humidity input Y-Axis

Fig. 16 (continued)

released in the process of defuzzification. And Table 6 and Table 7 respectively show the Arduino output control when rules from rule 1 to rule 12 are invoked then Result of greenhouse system using Fuzzy Logic. Table 8.

Simulation of fuzzy logic control shows the results performed on variable temperature and humidity as input variables obtained at 1:00 to 12:00 as in the Table 7 and can be seen in Fig. 17 below.

3.4 Result of Fuzzy Based Threat Model

Results of rule viewer for detecting threats at different levels of the layered IoT structure can be seen in the form of three-dimensional surface viewer as shown in Fig. 18a, b, c respectively. Surface viewer is plotting a graph of the data released in the process of defuzzification. Here we considered 3 major IoT perceptron layer attacks. The simulated predictions are found from testing rigorously with different values so that the attacks can be detected and prevented within safe timeframe.

3.5 Result with Confusion Matrix for Perceptron Layer Attack Detection

Confusion matrix is a k*k matrix where k defines specific parameters. Each cell (i; j) of confusion matrix represents the cases which actually belong to the attack category i and classified as attack category j. Performance measurement is done by comparing the actual class labels and the predicted class labels for each of the individual classifiers. This comparison is done to evaluate performance of the system model. The performances are measured using confusion matrix, classification accuracy, error rate,precision,F1-score and recall, kappa statistic to measure the improvement in insuring data security of the proposed system, following data show the predicted and actual dataset result after training and testing. It shows 4*4 confusion matrix [28]. Each row

Table 6 Arduino output control when rules from 1 to 12 are invoked

Temp & humidity	Low	Medium	High	Very high
High	Fan off, heater maximum, humidifier off (rule1)	Heater off, humidifier off, fan medium (rule2)	Heater off, humidifier off, Fan medium (rule3)	Heater off, humidifier off, Fan maximum (rule 4)
Medium	Fan off, heater medium, humidifier off (rule5)	Heater off, Fan off, humidifier off (rule6)	Humidifier off, heater off, fan medium (rule7)	Humidifier off, heater off, fan medium (rule8)
Low	Humidifier medium, heater maximum, fan off (rule9)	Heater off, fan off, humidifier maximum (rule10)	Heater off, humidifier medium, fan medium (rule11)	Heater off, humidifier off, fan maximum (rule12)

Table 7 Result of greenhouse system using Fuzzy Logic

Time	1	2	3	4	5	6	7	8	9	10	11	12
Temp (°C)	33.4	45	54.7	74	35.3	42.1	58.5	77.9	34.4	44	55.6	86.6
Humidity (%)	28.5	30.7	35.2	39.8	45.4	48.8	49.9	52.2	77	80.4	83.8	87.2

Table 8 Values of different attacks from simulation

	DDoS	MITM	Side Ch	Mal
DDoS	0.85555	0.021231	0.044444	0.082344
MITM	0.086678	0.811125	0.043456	0.058221
Side Ch	0.0654	0.042111	0.822222	0.062345
Malicious	0.08788	0.054321	0.052344	0.813187

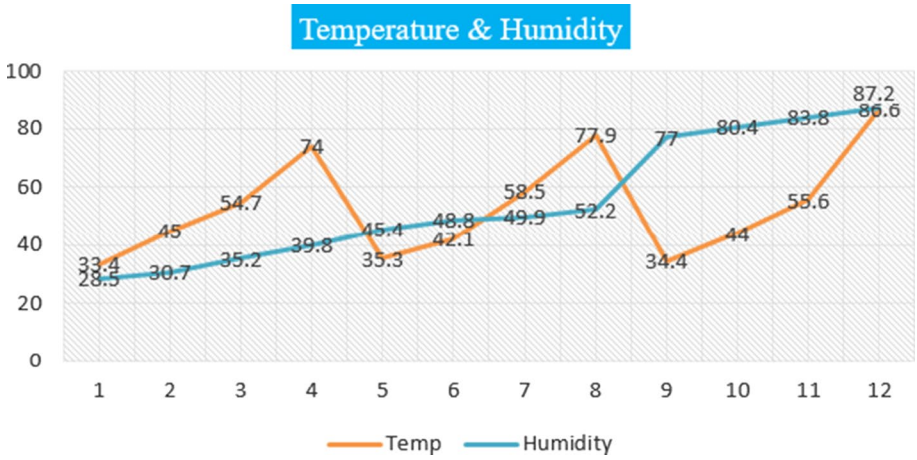


Fig. 17 Mesurement of temperature and humidity by line graph

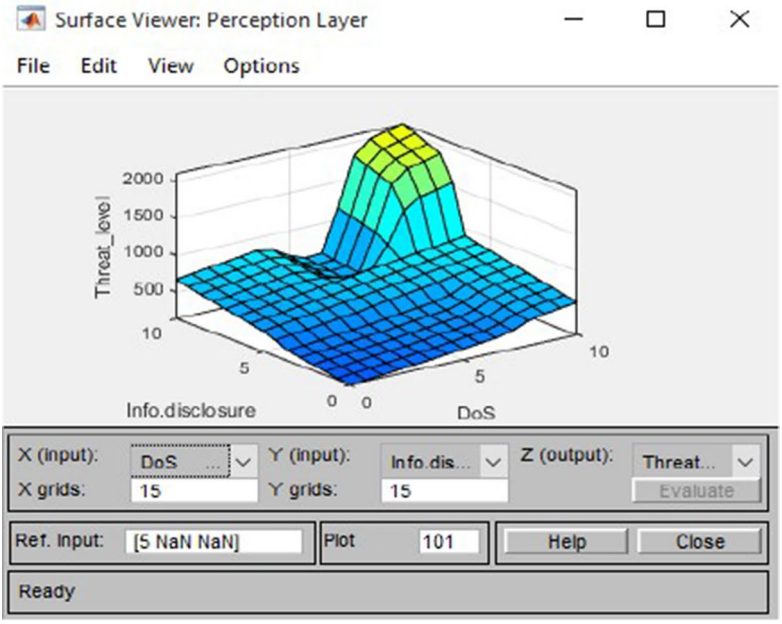
represents actual attack categories whereas each column represents one of the attack and the attacks are categorized by FIS Controller.

Primarily range of different attacks is defined like this-

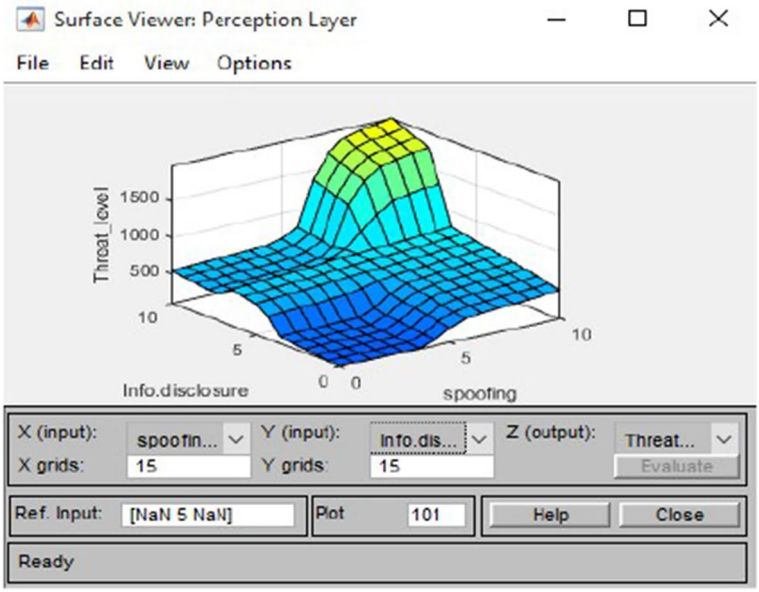
(i) Malicious Code Attack: 0.0–0.25
(ii) MITM: 0.25–0.5
(iii) Side Channel Attack: 0.5–0.75
(iv) DDoS: 0.75–1

After simulation, the value of four different types of attacks is summarized in Table 8.

The value of accuracy, precision, recall and F1 score is calculated from this confusion matrix which has been depicted at Table 9.

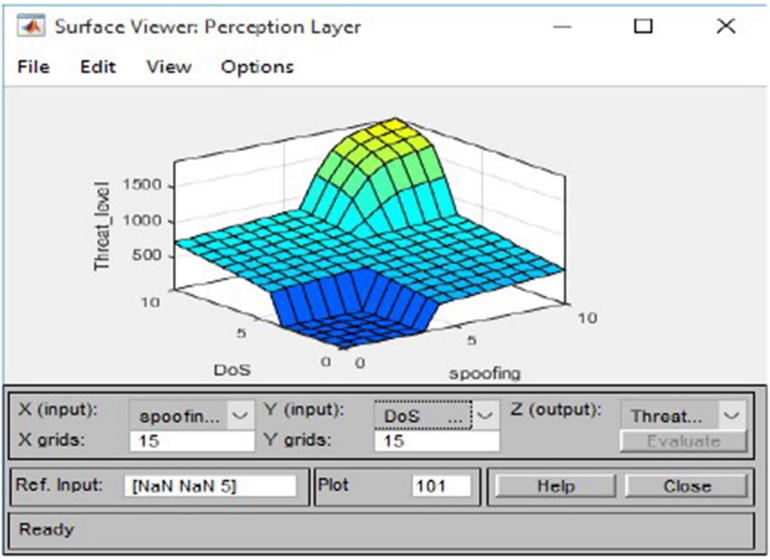


(a) Perception layer DoS attack surface view



(b) Perception layer Spoofing attack surface view

Fig. 18 **a** Perception layer DoS attack surface view **b** Perception layer Spoofing attack surface view **c** Perception layer Spoofing and Dos combined attack surface view



(c) Perception layer Spoofing and Dos combined attack surface view

Fig. 18 (continued)

Table 9 Result shown by confusion matrix measures

Accuracy (%)	Precision	Recall	F1-score
89.6	.84	.82	.83
92.62	.81	.86	.84
93.62	.83	.78	.81
89.26	.81	.83	.82

4 Conclusion

The developed system is appropriate for both large scale farming industry as well as small agriculture farm. The efficiency and accuracy of the system is more precise than labor-intensive systems. Human can observe whether the soil is wet or not. However, our system is able to measure the actual amount of moisture that is present in the soil as well as how much the greenhouse really need. It eradicates risk of human mistakes to maintain a greenhouse at a specific environmental condition and reduce the risk of attacks or hacks for data by ensuring securities at each layers of IoT by the confusion matrix and other mentioned measures. The implementation of this Smart Greenhouse is customizable. Again, the objective of this work is achieved to the extent that it has been shown through modeling and control by the use of fuzzy logic. The performance FLC was evaluated and has shown good control over output. This work also combines IoT technology for identifying data threat on a network transfer with fuzzy set, this approach improves the learning efficiency, enhance the predication accuracy, and proved to be feasible and effective automated greenhouse maintenance system compared to other recent proposed and available systems. At the same time, through the data collection module and presentation schema of data from

various sensors and security subsystem module achieve the cloud storing of data as well as the format conversion which comply with the protocol format data. As a result, it can provide traceability and durability of data for the quality and safety of customized indoor agriculture with the internet of things. Thus, farmers can really be benefitted if they adopt this "easy to use" system. They can control every environment parameters according to their needs with very simple mobile app. In the future, this system can be upgraded as an intelligent system by adding nano-robotic sensors and data processing modules, wherein the system predicts user's actions, rainfall patterns, time to harvest and other related features which will make the system independent of human operation(s). Moreover, this work can be improved to facilitate the extension of the mobile app, like adding alarms if a particular parameter is not controlled properly and the evaluation of the controller with ongoing crop production is mandatory, where a controlled group of plants will also be cultivated to provide with more rigorous testing of the proposed system.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11277-022-09528-x.

Funding This work was not supported by anyone. It was our own work.

Declarations

Conflicts of interest We have no conflicts of interest to disclose.

Availability of data and material The data supporting the findings of this study and its supplementary materials will be provided upon request to corresponding author(s).

Code availability The code of this article will be delivered upon request to corresponding author(s).

References

1. Modern Greenhouse. Retrieved from https://www.britannica.com/topic/greenhouse
2. Ravi K., Vishal J., Sumit K., (2016). IoT Based smart greenhouse. IEEE Region 10 Humanitarian Technology Conference (R10-HTC) DOI: https://doi.org/10.1109/R10-HTC.2016.7906846
3. Min, X., & Kuang, W. (2018). Study on the ecological farming control system based on the Internet of Things. *Wireless Personal Communication, 102*, 2955–2967. https://doi.org/10.1007/s11277-018-5318-0
4. Mayuri, A., & Sudhir, T. (2015). Internet of Things: Architecture, security issues and countermeasures. *International Journal of Computer Applications, 125*(14), 1–4.
5. Harbi, Y., Aliouat, Z., & Harous, S. (2019). A review of security in Internet of Things. *Wireless Personal Communication, 108*, 325–344. https://doi.org/10.1007/s11277-019-06405-y
6. Shicheng, Z., Shunkun, Y., Xiaodong, G., Yang, X., Tao, Z., & Yueliang, W. (2021). Survey of testing methods and testbed development concerning Internet of Things. *Wireless Personal Communication*. https://doi.org/10.1007/s11277-021-09124-5
7. Balaji, G. N., Nandhini, V., Mithra, S., Priya, N., & Naveena, R. (2018). IOT based smart crop monitoring in farm land. *Imperial Journal of Interdisciplinary Research (IJIR), 4*(1), 88–92.
8. Shouying, L., Shuyuan, L., Qijie, F., & Tengyue, Z. (2018). Research and implementation of a modern agricultural greenhouse cultivation system based on Internet of Things. *International Journal of Information Technology and Web Engineering, 13*(1), 39–49.
9. Özlem, A., & Ebubekir, E. (2018). The control of greenhouses based on fuzzy logic using wireless sensor networks. *International Journal of Computational Intelligence Systems, 12–1*, 190–203.
10. Shirsath, D. O., Punam, K., Rohini, M., Ashwini, K., & More, R. S. (2017). IOT based smart greenhouse automation using Arduino. *International Journal of Innovative Research in Computer Science & Technology (IJIRCST), 5*(2), 234–238.

11. Suma, N., Sandra, R., Saranya, S., Shanmugapriya, G., & Subhashri, R. (2017). IOT based smart agriculture monitoring system. *International Journal on Recent and Innovation Trends in Computing and Communication, 5*(2), 177–181.
12. Chiyurl Y., Miyoung H., Shin-Gak K., Juyoung P., Changkyu L. (2018). Implement smart farm with IoT technology. International Conference on Advanced Communications Technology (ICACT) ISBN 979–11–88428–01–4 ICACT.
13. Pavankumar, N., Arun, K., Vishwanath, H., Chaitra, N. K., Pavitra, H. K., Sushma, B. S., Sushmita, J., & Praveen, K. (2017). Arduino based automatic irrigation system using IoT. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology., 2*(3), 1–5.
14. Rodriguez, E. F., Kubota, C., Giacomelli, G. A., Tignor, M. E., Wilson, S. B., & McMahon, M. (2010). Dynamic modelling and simulation of greenhouse environments under several scenarios: A web based application. *Journal of Computers and Electronics in Agriculture, 70*(1), 105–116.
15. Dhamakale, S. D., & Patil, S. B. (2011). Fuzzy logic approach with microcontroller for climate controlling in green house. *International Journal on Emerging Technologies, 2*(1), 17–19.
16. Shukla, A. J., Panchal, M. V., & Patel, M. S. (2015). Intelligent greenhouse design based on Internet of Things (IoT). *International Journal of Emerging Trends in Electrical and Electronics, 11–2*, 57–61.
17. Almusaylim, Z. A., & Zaman, N. (2019). A review on smart home present state and challenges: linked to context-awareness Internet of Things (IoT). *Wireless Networks, 25*, 3193–3204. https://doi.org/10.1007/s11276-018-1712-5
18. Adhikary, A., Hasan, R., Hossen, K., et al. (2021). Performance analysis of transmission rate and decoding rate using minimum energy channel codes for nanoscale wireless communication. *Wireless Personal Communication, 118*, 1343–1364. https://doi.org/10.1007/s11277-021-08079-x
19. Srivastava, A., & Das, D. K. (2021). A comprehensive review on the application of Internet of Thing (IoT) in smart agriculture. *Wireless Personal Communication*. https://doi.org/10.1007/s11277-021-08970-7
20. Feng, X., Yan, F., & Liu, X. (2019). Study of wireless communication technologies on Internet of Things for precision agriculture. *Wireless Personal Communication, 108*, 1785–1802. https://doi.org/10.1007/s11277-019-06496-7
21. Harshit, B., Brij, B., & Nagesh, K. (2019). IOT: The current scenario and role of sensors involved in smart agriculture. *International Journal of Recent Technology and Engineering (IJRTE), 8*(4), 12011–23.
22. Marietta, J., & Chandra, M. (2020). Review on routing in Internet of Things. *Wireless Personal Communication, 111*, 209–233. https://doi.org/10.1007/s11277-019-06853-6
23. Fuzzy Logic Controller. Retrieved from https://www.mathworks.com help/fuzzy/ fuzzylogic controller.html
24. Xu, L. D., He, W., & Li, S. (2014). Internet of Things in Industries. *IEEE Transactions on Industrial Informatics, 10*(4), 2233–2244.
25. Fotohi, R., & Pakdel, H. (2021). A lightweight and scalable physical layer attack detection mechanism for the Internet of Things (IoT) using hybrid security schema. *Wireless Personal Communication, 119*, 3089–3106. https://doi.org/10.1007/s11277-021-08388-1
26. Mars, D., Mettali, G. S., & Lahmadi, A. (2019). Using information centric networking in Internet of Things: A survey. *Wireless Personal Communication, 105*, 87–103. https://doi.org/10.1007/s11277-018-6104-8
27. Pallavi, S., & Smruti, R. S. (2017). Internet of Things: Architectures, protocols, and applications. *Journal of Electrical and Computer Engineering, 26*, 2017.
28. Confusion Matrix. Retrieved from https://www.sciencedirect.com/topics /engineering/confusion -matrix

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Sultana Jahan Soheli has been serving as an Assistant Professor at Information and Communication Engineering department in Noakhali Science and Technology University (NSTU), Noakhali, Bangladesh since January 2018. She received her B.Sc. and M.Sc. Engineering degree in Information Technology from Jahangirnagar University, Dhaka, Bangladesh. Her research interests are currently focused on internet of things, machine learning, wireless communication and artificial intelligence.

Nusrat Jahan received her Bachelor's degree in Information and Communication Engineering from Noakhali Science and Technology University, Noakhali, Bangladesh in March 2020. At present, she is working as a Software Engineer in a Norway-based software company. Her areas of interest include IoT, Internet & Web Programming, Cloud Computing, and Machine Learning.

Md. Bipul Hossain is currently working as a Lecturer of Information and Communication Engineering at Noakhali Science and Technology University (NSTU), Noakhali, Bangladesh. He completed his B.Sc. and M.Sc. degrees in Information and Communication Engineering from Islamic University, Bangladesh. His current research interests are machine learning, computer vision, internet of things, wireless communication and optical communication.

Apurba Adhikary has been serving as an Assistant Professor at Information and Communication Engineering department in Noakhali Science and Technology University (NSTU), Noakhali, Bangladesh since January 2020. He received his B.Sc and M.Sc Engineering degrees in Electronics and Communication Engineering from Khulna University, Khulna, Bangladesh. His research interests are currently focused on wireless communication, internet of things, optical soliton and micro/nano-patterning of various materials.

Ashikur Rahman Khan working as a professor at Information and Communication Engineering department in Noakhali Science and Technology University, Noakhali, Bangladesh. He received his Phd from Pahang University Malaysia, M.Sc in engineering from Bangladesh University of Engineering and Technology, B.Sc. engineering from Rajshahi University of Engineering and Technology, (Mechanical Engineering). His research interests are currently focused on internet of things, machine learning and artificial intelligence.

M. Wahiduzzaman completed his undergraduate degree in Electronics & Computer Science and MSc in Telecommunication & Computer Network Engineering from London,UK. He also completed his PhD degree from University of Malaya, Malaysia. He serves as an Associate Professor in the Institute of Information Technology (IIT), Jahangirnagar University. Presently, he works as a Postdoctoral Reseach Fellow in a Australian Reseach Council (ARC) funded projects at Queensland University of Technology, Brisbane, Australia. His research interests are Mobile Cloud Computing, Vehicular cloud computing, Fog Computing, IoT, Microservices. Recently, he received the Elsevier JNCA best paper award in Paris, France. He visited and presented his works in many countries including India, Nepal, Thailand, Malaysia, Singapoure, Switzerland, Belgium, France, Germany, Australia, UK, and USA.

Authors and Affiliations

Sultana Jahan Soheli[1] · Nusrat Jahan[1] · Md. Bipul Hossain[1] · Apurba Adhikary[1,2] · Ashikur Rahman Khan[1] · M. Wahiduzzaman[3]

✉ Sultana Jahan Soheli
sjsoheli.ice@nstu.edu.bd; sohelisoha@gmail.com

✉ Apurba Adhikary
apurba@nstu.edu.bd; adhikary_apurba@yahoo.com

1 Department of Information and Communication Engineering, Noakhali Science and Technology University, Noakhali 3814, Bangladesh

2 Electronics and Communication Engineering Discipline, Khulna University, Khulna 9208, Bangladesh

3 Queensland University of Technology, Brisbane, Australia

View publication stats